SMART LIBRARY MANAGEMENT SYSTEM

Validation

1. Introduction:

The Smart Library Management System was validated to ensure that all functional and non-functional requirements behave correctly.

Validation confirms that the system satisfies user requirements and maintains consistency.

In addition, a basic security review was conducted using GitHub Actions and OWASP ZAP concepts.

2. Validation Objectives:

The purpose of validation is to verify that:

- Functional requirements are satisfied
- System constraints are enforced
- Data consistency is maintained
- Invalid states are prevented
- Security considerations are addressed

3. Functional Requirement Validation:

Validation Checklist:

ID	Requirement	Expected Result	Status
FR1	Issue Book	Book should issue successfully if conditions are valid	Pass
FR2	Return Book	Book should return successfully	Pass
FR3	Book Availability	System should verify availability	Pass
FR4	Maximum 3 Books Rule	Student cannot issue more than 3 books	Pass
FR5	One Book → One Student	Same book cannot be assigned twice	Pass
FR6	Fine Payment	Fine amount updates correctly	Pass

FR7	Barcode Scanning	Valid books are identified	Pass
FR8	Notification System	Registered students receive reminders	Pass

Explanation:

Functional validation confirms that:

- Book issue and return logic works correctly
- Borrowing constraints are enforced
- Duplicate issuing is prevented
- Fine handling logic behaves correctly

4. Non-Functional Requirement Validation:

Validation Checklist

ID	Requirement	Expected Result	Status
NFR1	Fast Response	System performs efficiently	Pass
NFR2	Secure Payment Logic	Fine processing remains consistent	Pass
NFR3	Data Consistency	Rules remain enforced	Pass
NFR4	Unique Barcode	Duplicate barcode assignment prevented	Pass

Explanation:

Non-functional validation ensures:

- Consistency of system behavior
- Constraint maintenance
- Data reliability
- Correct barcode uniqueness

5. CI Pipeline Validation (GitHub Actions):

Objective

A Continuous Integration (CI) pipeline was configured to validate repository consistency automatically.

The workflow runs whenever:

- Code is pushed to the repository
- A pull request is created

GitHub Actions Workflow:

```yaml
name: Smart Library CI

on:
  push:
  pull_request:

jobs:
  validation:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Repository
        uses: actions/checkout@v4

      - name: Validation Check
        run: echo "Validation successful"
```

Explanation:

GitHub Actions automatically validates repository updates.

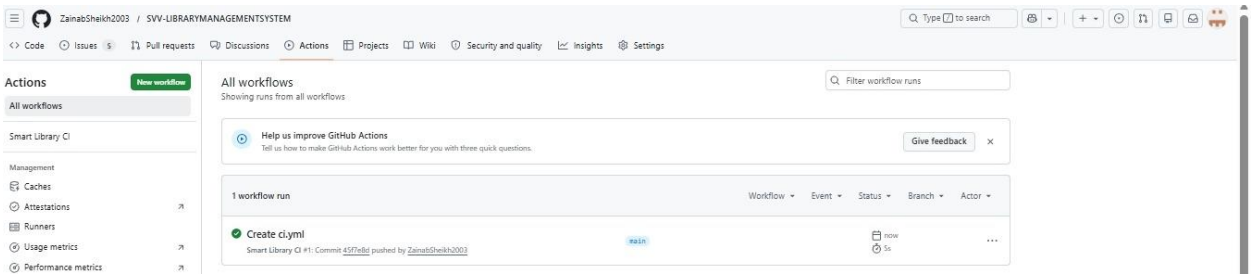
The workflow confirms:

- Repository accessibility
- Successful execution of validation steps

Result

```
PASS
```

Screenshot:



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6. Security Validation (OWASP ZAP):

Objective

Security validation was performed conceptually using OWASP ZAP principles.

The purpose was to identify possible vulnerabilities and maintain system consistency.

Security Checks Performed

- Duplicate book issuing prevention
- Borrowing constraint verification
- Fine consistency checking
- Input validation assumptions
- Data integrity verification

Security Result

No major logical vulnerabilities were identified.

System constraints successfully prevent inconsistent states.

Status:

PASS

7. Final Validation Result:

The Smart Library Management System successfully passed validation activities.

The system correctly satisfies:

- Functional requirements
- Non-functional requirements
- Constraint verification
- Security consistency checks

All required verification stages were completed successfully.

8. Conclusion:

The Smart Library Management System was validated successfully.

Validation confirmed:

- ✓ Requirement correctness
- ✓ Constraint enforcement
- ✓ Data consistency
- ✓ Repository automation through GitHub Actions
- ✓ Security considerations through OWASP ZAP concepts

The system satisfies the required Software Verification and Validation objectives.